ATM CASH WITHDRAWAL ANALYSIS

Data Analytics Project

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20,000-record project dataset

1. Business Problem

Why ATM cash analytics matters

CASH AVAILABILITY

Run-out risk

Maintain enough cash without unnecessary refill trips.

SERVICE QUALITY

Failed transactions

Failures and downtime can affect customer experience.

RISK

Suspicious activity

Large, night-time and high-frequency patterns can require review.

Existing monitoring is often reactive:

- Cash balance and withdrawals
- Number of transactions
- Refilling and downtime
- Failed transactions

Project goal

Turn transaction-level data into a management dashboard for cash, service and fraud-related decisions.

2. Project Objectives

01	**Analyze cash withdrawals**	Measure withdrawal volume and ATM demand.
02	**Monitor ATM performance**	Track failures, downtime, complaints and health.
03	**Support refill decisions**	Identify ATMs that may require replenishment.
04	**Monitor suspicious activity**	Surface large, night-time and high-frequency patterns.

3. Raw Data & Project Structure

The starting point for the analytics workflow

DATA VOLUME

20,000

Records prepared for the ATM analysis project.

DATABASE

SQL

Used as the database layer before BI analysis.

BI TOOL

Power BI

Used for cleaning, DAX and visualization.

OUTPUT

2 dashboards

Performance + smart/fraud monitoring.

Key data areas

- ATM and transaction identifiers
- Date / day / weekend information
- Opening, loaded, withdrawal and closing cash
- Failed transactions and service indicators
- Refill, health and complaint indicators
- Fraud indicators and fraud reasons

4. SQL Stage

Database workflow used before Power BI

Load data 1

Imported the prepared ATM dataset into SQL.

Run querie 2

Worked with the project data through SQL.

Validate fie 3

Checked required columns for analysis.

Connect to 4

Used SQL data as the BI source.

Result: structured database layer ready for BI analysis.

5. Power BI Data Cleaning

Validation performed before dashboard creation

DUPLICATES

Checked duplicates and avoided blindly deleting rows when row count changed significantly.

NULL / EMPTY

Checked for null or empty values; the Date table shown had 0% empty.

DATA TYPES

Validated date and numeric types, including cash amount fields.

NEGATIVE VALUES

Checked cash-related fields and closing-cash logic before filtering.

DATES

Validated Date values and used day/weekend fields.

FRAUD FIELDS

Checked Fraud_Flag and Fraud_Reason without inventing missing categories.

Cleaning principle: validate first → filter only when the business rule supports it.

6. DAX Measures

Turning raw columns into business KPIs

TOTAL CASH WITHDRAWN

SUM of withdrawal values

Example:
Total Cash Withdrawn =
SUM('ATM Table'[Total_Withdrawal])

FAILED TRANSACTION %

Failed Transactions ÷ Total Transactions

Example:
Failed Transaction % =
DIVIDE([Failed Transactions],
[Total Transactions], 0)

Other measures created/used

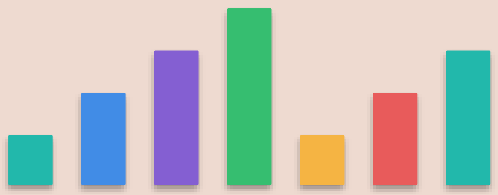
- Total Transactions
- Failed Transactions
- Average Health Score
- ATMs Requiring Refill
- Total Refill Amount / Cost
- Average Downtime
- Customer Complaints
- Suspicious / Large / Night / High-Frequency transactions

7. Dashboard 1 — ATM Performance & Cash Management

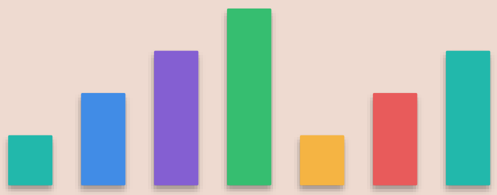
Management view of cash, service quality and refill needs

Total Cash Withdrawn	Failed Transaction %	Avg Health Score	ATMs Requiring Refill
₹2.81M	2.35%	76.4	142

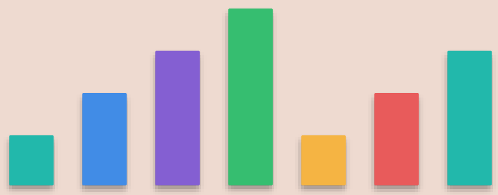
Cash Withdrawal Trend



ATM Cash / Refill Status



Top ATMs by Withdrawal



Purpose: identify cash-demand, service and refill priorities.

8. Dashboard 2 — Smart Monitoring & Fraud Analysis

Risk-focused view of unusual withdrawal patterns

Large Withdrawals

579

Night Transactions

3,413

High-Frequency Transactions

5,409

SUSPICIOUS TRANSACTIONS BY ATM

Compare ATMs with higher suspicious-transaction counts and focus review where risk is concentrated.

SUSPICIOUS TRANSACTIONS BY REASON

Break down suspicious activity by reasons such as large withdrawal, night-time, multiple withdrawals and distant ATM activity.

Purpose: prioritize transactions and ATMs for investigation instead of reviewing every record manually.

9. Key Insights & Business Value

CASH DEMAND

Identify ATMs with higher withdrawal activity and potential refill pressure.

SERVICE QUALITY

Monitor failure rate, downtime and health score to prioritize attention.

REFILL PLANNING

Use refill indicators to reduce unnecessary trips and avoid shortages.

FRAUD REVIEW

Use transaction patterns to flag unusual behavior for investigation.

Important: suspicious indicators support review; they do not prove fraud by themselves.

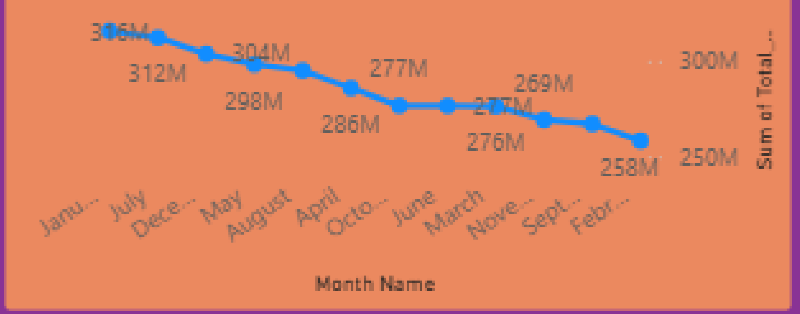
10. Challenges I Faced & How I Solved Them

1	**Data quality**	Checked duplicates, nulls, date errors, data types and negative-value logic.
2	**Row-count changes**	Avoided deleting records blindly when filters caused a significant drop.
3	**Fraud categories**	Worked with the categories actually present instead of inventing missing ones.
4	**Power BI learning**	Built step-by-step: cleaning → measures → KPI cards → charts → formatting.

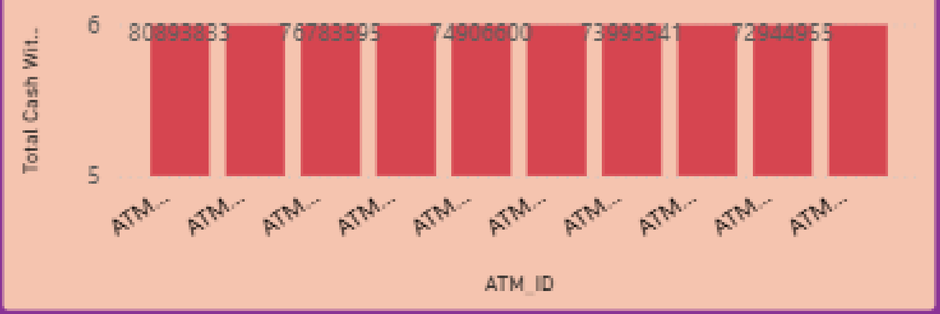
ATM PERFORMANCE & CASH MANAGEMENT

Total Cash Withdrawal	Total Transition	Failed Transaction %	Avg Health Score	ATM's Requiring refil
3.44bn	2.8M	0.02	81.92	50
▲ 0.0%	▲ 0.0%	1.95% ▼	82 ▲	50 ▲
vs Last period	vs Last period	vs Last period	vs Last period	vs Last period

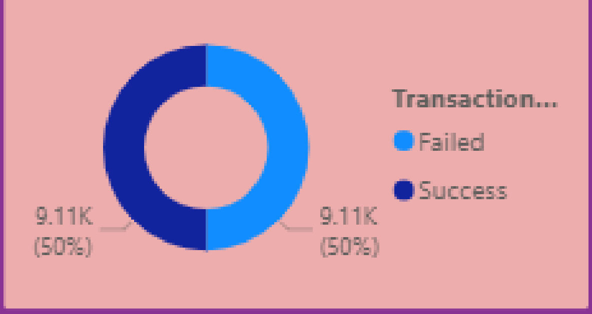
Total Cash Withdrawn Over Time



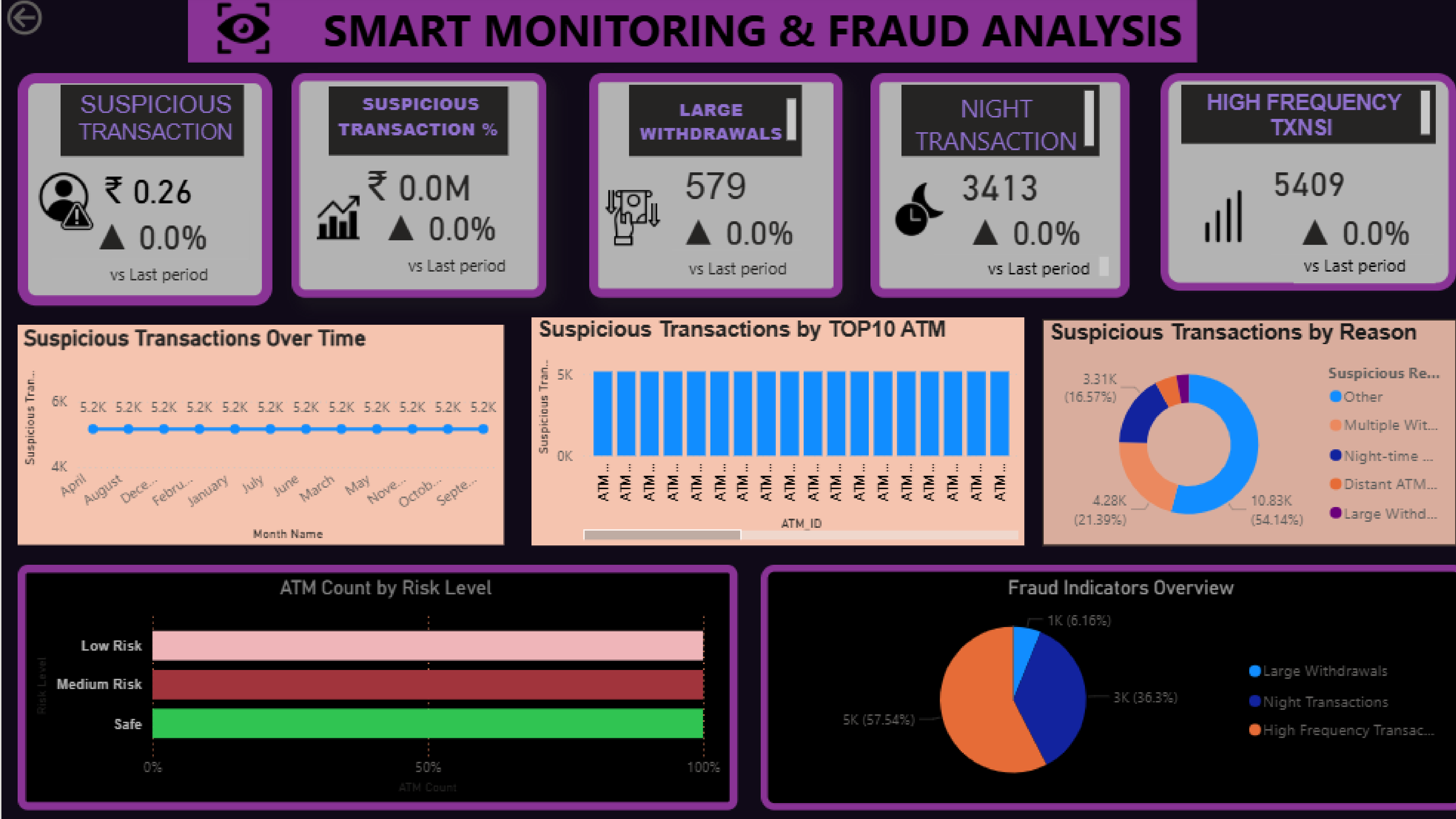
Top 10 ATMs by Cash Withdrawn



Transaction Status Distribution



Total Refill Amount	Total Refill Cost	Avg Downtime (Min)	Total Complaints
₹ 1.95bn	13M	11.98	18K
▲ 9.1%	▲ 100.0%	▼ 12.6%	▼ 8.4%


SMART MONITORING & FRAUD ANALYSIS
SUSPICIOUS TRANSACTION
₹ 0.26
▲ 0.0%
vs Last period
SUSPICIOUS TRANSACTION %
₹ 0.0M
▲ 0.0%
vs Last period
LARGE WITHDRAWALS
579
▲ 0.0%
vs Last period
NIGHT TRANSACTION
3413
▲ 0.0%
vs Last period
HIGH FREQUENCY TXNSI
5409
▲ 0.0%
vs Last period
Suspicious Transactions Over Time
Suspicious Tran...
6K
4K
5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K 5.2K
April August Dece... Febru... January July June March May Nove... Octob... Septe...
Month Name
Suspicious Transactions by TOP10 ATM
Suspicious Tran...
5K
0K
ATM ...
ATM_ID
Suspicious Transactions by Reason
3.31K (16.57%)
4.28K (21.39%)
10.83K (54.14%)
Suspicious Re...
Other
Multiple Wit...
Night-time ...
Distant ATM...
Large Withd...
ATM Count by Risk Level
Low Risk
Medium Risk
Safe
Risk Level
0%
50%
100%
ATM Count
Fraud Indicators Overview
1K (6.16%)
3K (36.3%)
5K (57.54%)
Large Withdrawals
Night Transactions
High Frequency Transac...

11. End-to-End Workflow & Conclusion

RAW DATA → SQL → POWER BI CLEANING → DAX MEASURES → DASHBOARD

Final outcome

- Cleaned and structured ATM analytics data
- SQL-backed workflow
- Reusable DAX measures for KPIs
- Two Power BI dashboards
- Decision-support view for ATM operations

THANK YOU

Questions & Discussion